Mapping Treatment Evolution and Adverse Events in Cystic Fibrosis: A Corpus-Based Text-Mining and Network Analysis of PubMed Literature

by

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Abstract

Background

Cystic Fibrosis (CF) has shifted from a fatal childhood disease to a chronic condition, with today's median survival above 60 years old in high-income countries, reflecting recent advances from symptomatic management to targeted therapies with CFTR modulators. As CF scientific literature grows, tracking how treatment and adverse-event terminology have evolved becomes increasingly difficult through traditional review methods.

Aim

This study aimed to use text mining and network analysis to characterise trends in treatment and adverse-event terminology across 55 years of CF literature (1970-2024).

Methods

Abstracts were retrieved from PubMed across 11 subsets and 11 quinquennial periods, yielding 41,222 unique articles after deduplication. *N*-gram frequency and weighted log-odds analysis identified period-distinctive terminology. Vocabulary richness was quantified using the Gini coefficient, Yule's *K*, and hapax legomena. Co-occurrence networks mapped treatment and adverse-event associations.

Results

From 1970-1974 to 2020-2024, the corpus grew from 502 articles to nearly 7,300. Vocabulary became more concentrated (Gini coefficient: 0.479 to 0.873) while lexical diversity increased too (Yule's *K*: 37.33 to 15.60), reflecting growth of specialised vocabulary and dominance of frequently used terms. Terminology shifted from general clinical terms and biological language to CFTR modulator and outcome-specific terms by 2020-2024. Network analysis identified a CFTR modulator cluster linked to adverse-event terms and terms describing CFTR mechanism terminology, alongside other clusters related to infection and mental health. CFTR modulator and mental health co-occurrence was absent prior to 2015-2019 but appeared in 2020-2024, though it was rare (relative to each term's individual frequency).

Conclusions

These findings show that large-scale text mining can discover longitudinal treatment and adverse-event patterns that are difficult to detect manually, while also highlighting a developing modulator-related mental-health association, thus warranting further clinical investigation.

Keywords: Text Mining; PubMed; Cystic Fibrosis; Adverse Events; Network Analysis; Temporal Analysis

1. Introduction

Cystic Fibrosis (CF) is a genetic condition resulting from a faulty CFTR gene. This gene, and the protein it produces, normally regulates the movement of salt and water across epithelial cell membranes, such as those lining the lungs and gut. When the gene and protein are faulty, mucus throughout the body becomes abnormally thick, affecting several organs simultaneously. The lungs are faced with mucus that blocks the airways, leading to respiratory infections, organ damage, and a shortened life expectancy (Shteinberg, 2021).

How CF is managed clinically has changed dramatically over time. In the 1960s, CF was characterized as a fatal childhood illness, and few patients lived past the age of 5 (Scotet et al., 2020). Treatment has since advanced from symptomatic management to CFTR modulators, and median survival for a child born in 2026 in high-income countries is now estimated at 66 years (UK Cystic Fibrosis Registry, 2025). Despite progress, people with CF still rely on multiple therapies long term; many times, a lifelong basis. Therefore, deeper understanding of long-term disease management, treatment, and adverse outcomes remains essential to improving survival and quality of life.

Text mining is the automated process of extracting meaningful patterns and information from large quantities of unstructured text (Silge & Robinson, 2017). Using text mining methods like tokenisation, frequency analysis, and network analysis allows a systematic, efficient way of identifying patterns that would otherwise be difficult to detect through non-automated methods.

PubMed is a freely accessible database holding over 40 million citations and abstracts from scientific and medical research (National Library of Medicine, 2025). Because it provides titles, abstracts, medical subject headings, and publication dates in a consistent format, this metadata can be used for systematic text mining and analysing trends in the CF literature. English accounts for approximately 95.7% of scientific journal publications, which helps keep the language used across this literature relatively simple and consistent for text mining purposes (Karcioglu et al., 2025).

Biomedical databases like PubMed have ever-increasing literature. MEDLINE, PubMed's database, had already exceeded 23 million indexed articles by 2017 (Renganathan, 2017), making manual extraction of knowledge time-consuming. Text mining allows vocabulary diversity to be quantified and patterns in treatment and adverse event related language to be found across literature (Renganathan, 2017; Harpaz et al., 2014). Network analysis builds on this by representing quantified terms as nodes and their co-occurrence frequency as edges, illustrating how the relationships between treatments and adverse events have developed over time (Al-Aamri et al., 2019).

Since the CFTR gene was identified in 1989, CF treatment has moved away from purely managing symptoms (e.g. mucoactive therapies such as dornase alfa and long-term nebulised antibiotics such as tobramycin) and to targeted molecular therapies. This shift introduced CFTR modulators and combination therapies, such as elexacaftor/tezacaftor/ivacaftor (Lopes-Pacheco, 2020; Scotet et al., 2020), which have been available within routine care over the past decade.

Despite this shift to CFTR modulator therapy, relatively little work appears to have tracked how CF-related treatment and adverse-event terminology has evolved across the wider biomedical literature. Existing reviews of CF treatment and safety tend to focus on a single therapy or a specific outcome at a specific point in time, rather than tracing how the language of treatment and adverse events has changed over time. Text mining and network analysis offers a way to address this gap by systematically extracting and quantifying terminology across thousands of abstracts. Longitudinal patterns can then be identified in cases where it otherwise would be difficult to detect through manual review alone.

Therefore, this study aims to characterise how treatment and adverse-event-related terminology in CF literature has changed between 1970 and 2024, using text mining and network analysis. Specifically, the study's objectives are:

1. To identify terminology most distinctively associated with each period of CF literature, using *n*-gram frequency and weighted log-odds analysis
2. To quantify how vocabulary diversity and concentration have changed across the study period
3. To map co-occurrence relationships between treatment and adverse-event-related terms using network analysis, to visualise how these associations have evolved
4. To examine the extent of co-occurrence between CFTR modulator therapy and mental-health-related terminology, given that prior literature has noted a possible association between the two (Piehler et al., 2023)

Understanding these trends is important. First, as new modulator and combination therapies continue to enter clinical use, an evidence-based understanding of how their associated safety language has developed over time could be important for ongoing drug safety. Additionally, demonstrating that text mining can reveal these patterns does support its use as a scalable, reproducible method as opposed to traditional literature review methods.

2. Methods

2.1. Dataset

The dataset used (PubMed) is a publicly available data source (National Library of Medicine, 2025). All text mining analysis used the programming language R. To obtain the corpus, the `pubmedR` package was applied programmatically (Cuccurullo, 2017). PubMed has public biomedical literature indexed with titles, abstracts, and publication dates, making it ideal for text mining.

2.2. Search Strategy

Subset Construction
Eleven subsets were constructed to address different attributes of CF literature. A shared base search string labelled "`parent`" was used across all subsets to capture core CF and CFTR modulator terminology, with each subset extending this string to target its specific area of

interest. Boolean operators (OR, AND) were used to add additional subset-specific criteria to the base-term set across the [tiab] field tag, meaning title, and abstract:

1. **Parent** - searched for CF-related terms
    - `"cystic fibrosis"[tiab] OR "CFTR modul*"[tiab] OR mucoviscidosis[tiab]`
        - "cystic fibrosis"[tiab] was chosen as the specific disease of interest
        - "CFTR modul*"[tiab] was chosen because it is a CF-specific treatment
2. **Lung**
    - `lung[tiab] OR airway*[tiab] OR pulmonary[tiab] OR respirat*[tiab]`
3. **Liver**
    - `liver[tiab] OR hepat*[tiab]`
4. **Pancreas**
    - `pancrea*[tiab]`
5. **Treatment**
    - `treatment[tiab] OR therap*[tiab] OR manag*[tiab] OR care[tiab]`
6. **Life Expectancy**
    - `"life expectancy"[tiab] OR surviv*[tiab] OR mortalit*[tiab]`
7. **Modulators** - searched for CFTR modulator therapy-related terms
    - `ivacaftor[tiab] OR lumacaftor[tiab] OR elexacaftor[tiab] OR tezacaftor[tiab] OR kaftrio[tiab] OR orkambi[tiab] OR trikafta[tiab] OR symkevi[tiab] OR symdeko[tiab] OR kalydeco[tiab] OR vanzacaftor[tiab] OR alyftrek[tiab]`
8. **Aminoglycosides** - searched for aminoglycoside-related terms as common long-term treatments in CF
    - `aminoglycoside*[tiab] OR tobramycin[tiab] OR amikacin[tiab]`
9. **Enzyme/Dornase** - searched for mucoactive agent-related terms as common long-term treatments in CF
    - `dornase*[tiab] OR DNase[tiab] OR pulmozyme[tiab] OR hypertonic*[tiab] OR mannitol[tiab] OR mucoactiv*[tiab] OR mucolytic[tiab]`
10. **Adverse Event**
    - `"adverse event*"[tiab] OR "side effect*"[tiab] OR toxicit*[tiab] OR "adverse effect*"[tiab] OR safety[tiab]`
11. **Mental Health**
    - `"mental health"[tiab] OR depression[tiab] OR anxiety[tiab]`

Quotation marks (" ") were used to search for compound words as singular terms. Right-hand truncation (*) was applied to capture different forms of a word (e.g. modul* captures modulator and modulators).

This approach allowed targeted investigation of specific treatments and adverse event terms, but did create overlapping subsets requiring deduplication (Section 2.3).

Filters
All subsets applied two filters:

- `english[lang]`- filtered for English-only publications
- `humans[MeSH Terms]`- filtered for studies only involving humans

Date Range
The publication date range was constrained to 01/01/1970 to 31/12/2024 using the [dp] field tag. It was applied at the end of each query along with the base-term and subset-specific terms.

- `"1970/01/01"[dp] : "2024/12/31"[dp]`

CF was first described in 1938 (Scotet et al., 2020), and the earliest PubMed-indexed abstract containing the term "cystic fibrosis" was in 1956. However, the number of abstracts before 1970 was negligible. Therefore, 01/01/1970 was chosen as the start date for publications undergoing text mining analysis. To allow 11 consistent 5-year intervals (quinquennial periods), the end date of 31/12/2024 was chosen, thereby excluding partial-year data from 2025 and 2026 for consistent temporal trend analysis.

Excluding MeSH Terms
Medical Subject Headings (MeSH) were used only as a filter to restrict retrieval to human studies (`humans[MeSH Terms]`), thereby excluding animal research. MeSH terms were not used to identify disease or treatment concepts within the search strategy itself, as they had inconsistent indexing coverage in pre-1970s literature and identify publications by concept rather than by explicit wording in the text, which would have defeated the purpose of using free-text search terms and n-gram-based text mining and network analysis.

2.3. Corpus Retrieval and Deduplication

For each of the 11 subsets, queries were run separately for each quinquennial period. A total of 121 queries were run, with the results retrieved using `pubmedR::pmApiRequest()` after confirming the total counts with `pubmedR::pmQueryTotalCount()`. All results included PMID, title, abstract, and publication date, labelled with the subset name and associated time period.

As mentioned in Section 2.2, deduplication was required. The 11 subsets were constructed with overlapping inclusion criteria. For example, a publication discussing CFTR modulators and lung outcomes might have been retrieved by both the "modulators" and "lung" subset. The retrieved records were consequently then deduplicated based on PMID, leaving only one record per unique article:

- **Total records retrieved across all subsets and periods (pre-deduplication):** *96,525*
- **Unique articles remaining after deduplication:** *41,222*

All subsequent pre-processing and analysis used the deduplicated corpus.

2.4. Pre-processing

Two pre-processing pipelines were created using the deduplicated corpus: a lemmatised pipeline for unigram analysis (single-words) and a raw pipeline for *n*-gram (multi-word) analysis, specifically bigrams and trigrams. The pipelines were kept separate because *n*-gram analysis depends on preserving the original word order and sentence context. As for unigram frequency analysis, it benefits from the dimension reduction that lemmatisation and stopword removal provide.

Lemmatised pipeline (unigram analysis):

1. **Lowercasing & Punctuation removal** - all text was converted to lowercase and punctuation was removed to reduce dimensionality
2. **Tokenisation** - text was cut up into smaller units (tokens) using `tidytext::unnest_tokens()` (Silge & Robinson, 2017)
3. **Stopword removal** - tokens were matched to `tidytext`'s `stop_words` lexicon, with a supplemental, manually constructed list of other noisy terms identified within the corpus during manual exploration
4. **Lemmatisation** - remaining tokens were reduced to their base form using the `textstem` package (Rinker, 2018)

Raw pipeline (n-gram analysis):

1. **Lowercasing & Punctuation removal only**
    a. *n*-gram analysis benefits from preserving the original word-order-context. Therefore, only lowercasing and punctuation removal were applied for bigram/trigram analysis
        i. Tokenisation, stopword removal, and lemmatisation were omitted to avoid losing contextual integrity of words or phrases, which aligns with research concerning *n*-gram frequency analysis (Yip et al., 2010)
2. **US vs. UK spelling standardisation**
    a. US and UK spelling variations (e.g. tumor vs tumour) were found and standardised during exploratory analysis to prevent different spellings from being treated as distinct terms

2.5. Term Identification and Frequency Analysis

Treatment and adverse-event-related terms were identified using n-gram frequency analysis. Unigrams, bigrams, and trigrams were extracted from the raw pre-processed corpus and counted within each quinquennial period. To keep only recurring, meaningful phrases and exclude low-frequency noise, a minimum frequency threshold of ≥10 occurrences per period was applied to bigrams, and ≥5 occurrences per period to trigrams. These thresholds were then tested at several levels before being finalising (Yip et al., 2010). Unigrams were not thresholded at this stage because the complete frequency distribution was needed separately for vocabulary richness analysis (Section 2.7).

Weighted log-odds ratios (Monroe et al., 2008) were then calculated using the `tidylo` package to identify terms disproportionately associated with a given period relative to the other terms (Pojanapunya et al., 2018). Instead of a simple log-odds approach, a weighted log-odds approach was used because an initial attempt at simple log-odds calculations produced unstable results for periods with a small total corpus size. This led to extreme scores being assigned to generic, non-essential terms in the earliest periods. This was likely a byproduct of small-sample volatility, not actual linguistic distinctiveness. The weighted approach corrects for this by accounting for the quantity of data behind each estimate.

Several rounds of filtering were then applied to the log-odds output to uncover meaningful treatment and adverse event-related terms from language that was statistically prevalent but not necessarily analytically useful. First, exact-match disease-name phrases like "cystic fibrosis" and "cf patients" were excluded. Second, abbreviation-definition remnants were removed. A common abbreviation for “people with cystic fibrosis” is “PWCF”, so when an abstract spelt this out fully before using the abbreviation (e.g. “cystic fibrosis (PWCF)”), punctuation removal caused this to collapse into what seemed to be two separate terms appearing together (“cystic fibrosis pwcf”), when in reality, it’s capturing the tail end of this abbreviation pattern. Third, stopwords and generic disease terms such as "patients" or "disease" were removed. Fourth, systematic review and methodology terms, such as "confidence interval" or "risk of bias" were removed, since these aren't specific to CF. Lastly, fragmented concept variants were combined. Small differences in symbols, spacing, and the effects of punctuation removal meant that case like the F508del CFTR mutation and the CFTR gene/protein itself showed up as more than ten different-looking terms. These were subsequently all mapped to one standard form before being re-counted, and log-odds were recalculated using the combined counts.

High-frequency but analytically unimportant terms dominated initial weighted log-odds results (mainly variants of "cystic fibrosis" and "CF" itself, along with generic clinical terms). These obscured the treatment and adverse-event-specific terms that were of interest. The filtering process in Section 2.5 was therefore applied across the unigram, bigram, and trigram results before final term rankings were made.

2.6. Vocabulary Richness Analysis

Linguistic diversity was quantified using 3 metrics, calculated from the lemmatised, unigram-level corpus:

1. **Gini Coefficient** - measures inequality in word-frequency distribution, originally developed for income inequality (Gini, 1912; Ceriani and Verme, 2011) and then applied to vocabulary distribution analysis (Popescu et al., 2009); values range from 0 (perfect equality) to 1 (perfect inequality)
2. **Yule's *K*** - assesses lexical diversity and repetition (Tanaka-Ishii et al., 2015); lower values indicate greater diversity, higher values indicate greater repetitiveness
3. **Hapax Legomena** - the count of words occurring exactly once within a period's corpus (Tweedie & Baayen, 1998); relative higher values indicate richer vocabulary

These metrics were calculated using the full, non-thresholded unigram frequency distribution since all of the metrics depend on the full distribution of word frequencies.

To examine if periods with more statistically distinct terms also had a more concentrated vocabulary, the median weighted log-odds score of each period's top 20 most distinctive terms (calculated separately for unigrams, bigrams, and trigrams) was correlated against that period's Gini coefficient. Spearman's rank correlation was used rather than Pearson's, since only a monotonic relationship between the two measures could be assumed.

2.7. Temporal Analysis

All analyses (term frequency, log-odds ratio, vocabulary richness, and network analysis) were conducted separately within each of the 11 quinquennial periods (1970-2024), rather than on the full corpus at once, to enable longitudinal comparison of how treatment and adverse-event-related terminology evolved over the studied timeframe.

2.8. Network Analysis

Co-occurrence networks were constructed to map associations between treatment and adverse-event-related terminology, with nodes representing terms and edges representing their co-occurrence within the same publication (Newman, 2010).

Base vocabulary
Two base term lists were built from the search vocabulary in Section 2.2: a treatment list (CFTR modulators/brand names, aminoglycosides, and mucolytic/enzyme therapies) and an adverse-event list (general adverse-event and safety terms, plus mental-health terms). Each term was checked against the corpus to confirm it appeared before being included.

Data-driven expansion
The base vocabulary was then supplemented with relevant terms that hadn't been anticipated at the search design stage. A wide net of period-distinctive candidate terms (the top 50 unigrams, bigrams, and trigrams per period by weighted log-odds) were tested for co-occurrence with the base vocabulary across the full corpus. Candidate terms were kept only when they co-occurred with at least one base term ≥ 30 times. There were several new, relevant additions that weren't a part of the original search strategy, including a specific pathogen (*Pseudomonas aeruginosa*) and standard pulmonary outcome measures (forced expiratory volume in 1 second, FEV1), along with several others.

Vocabulary Refinement
The combined vocabulary was then refined even further:

- **Stopword phrases removed** - any phrase containing a stopword was dropped, which is a stricter rule than the one used for the general frequency analysis. This is because a meaningful network node needs to represent a complete concept
- **Fragmented concept variants combined** - extending the approach from Section 2.6 to also cover plural/singular forms (since lemmatisation hadn't been applied to the raw n-gram pipeline)

- **Generic scientific-writing terms excluded** - words like "significant" and "reported". These were found through manual review of the network graph, since they show up in almost every abstract regardless of topic, so they don't indicate an actual treatment or adverse-event-specific association

Co-occurrence matrix and thresholding
For each period, a matrix was built showing which terms appeared in which abstracts. From this, co-occurrence counts were calculated, showing how often each pair of terms showed up together in the same abstract. An initial threshold of 10 produced an unreadable network, so this was increased to 30 co-occurrences. Edges were also limited to those involving at least one base term (treatment or adverse-event), since the goal was to map treatment to adverse-event associations specifically. Finally, the network was narrowed down to only its largest connected group of terms for visualisation using the `quanteda` package (Benoit et al., 2018) to create the document-term matrix and obtain co-occurrence counts, and the `igraph` package (Csárdi and Nepusz, 2006) to create the network graph and extract its largest connected component.

Visualisation
An initial approach using an algorithm to automatically detect clusters (Blondel et al., 2008) was explored, but a more directed approach was used instead. Terms were manually classified as “treatment”, “adverse event”, or “associated term”, based on whether they were in the base vocabulary. Node size reflected how many connections each term had. This gave a clearer view of what the study aims to show (illustrating treatment-to-adverse-event associations) than an automatic clustering algorithm would, since it wouldn't necessarily separate terms into these two categories on its own. Prior literature has also noted a possible link between CFTR modulators and mental health effects (Piehler et al., 2023); direct co-occurrence edges between CFTR-modulator treatment terms and mental-health adverse-event terms were therefore checked for, though none met the co-occurrence threshold used to construct the main network. The rest of the network was coloured by time period.

Results

3.1 Corpus Composition

After retrieving PubMed records across all 11 subsets and 11 quinquennial periods, 94,597 records were initially returned. Because the other ten subsets were built by adding extra criteria on top of one shared base search (parent), many records were the same publication retrieved more than once, but simply under different subsets. After removing these duplicates using each article's PubMed ID, 41,222 unique articles remained.

The number of unique articles per period grew steadily across the study period, from 502 articles in 1970-1974 to 7,314 by 2020-2024 (Table 1). This growth reflects the overall expansion of CF-related research literature over time and is the reason this analysis uses per-article rates (rather than raw counts) when comparing term usage across periods. Without this adjustment, later

periods would appear to discuss a term more simply because more articles were published, not because the topic became genuinely more prominent.

Table 1. Number of unique articles retrieved per quinquennial period following deduplication (total N = 41,222).

Number of Articles Retrieved per Quinquennial Period

After deduplication on PMID

Period	Articles (n)	% of Total
1970–1974	502	1.2
1975–1979	913	2.2
1980–1984	1,269	3.1
1985–1989	2,066	5.0
1990–1994	3,134	7.6
1995–1999	3,652	8.9
2000–2004	4,165	10.1
2005–2009	4,969	12.1
2010–2014	6,285	15.2
2015–2019	6,953	16.9
2020–2024	7,314	17.7
Total	**41,222**	**100.0**

3.2 Term Frequency and Period-Specific Terminology

Weighted log-odds analysis (Section 2.5) was used to find the unigrams, bigrams, and trigrams most strongly linked to each specific time period, after removing disease-name terms, systematic-review terms, and abbreviation byproducts (Section 2.6). Figures 1-3 show the ten most distinctive terms for each of the 11 periods.

These period-specific terms tell a clear clinical story over time. In the earliest periods, the most distinctive terms are generic clinical words like "lung", "treatment", and "function" alongside genetic and biological terms that emerged after the CFTR gene was discovered in 1989 ("transmembrane", "gene", "cftr", etc.). Through the 1990s and 2000s, terminology becomes more focused on CFTR-specific genetics and CF-associated infections("cftr", "biofilm", "burkholderia"). From 2010-2014 onward, the most distinctive terms shift clearly toward named CFTR modulator drugs (ivacaftor, lumacaftor), and by 2020-2024, toward combination CFTR modulator therapies, standard lung function measures (forced expiratory volume in one second, FEV1), and terms related to COVID-19.

No bigrams or trigrams met the minimum frequency threshold in the earliest period, 1970-1974, so this period does not appear in Figures 2 and 3. This is simply because the corpus was very small in this period.

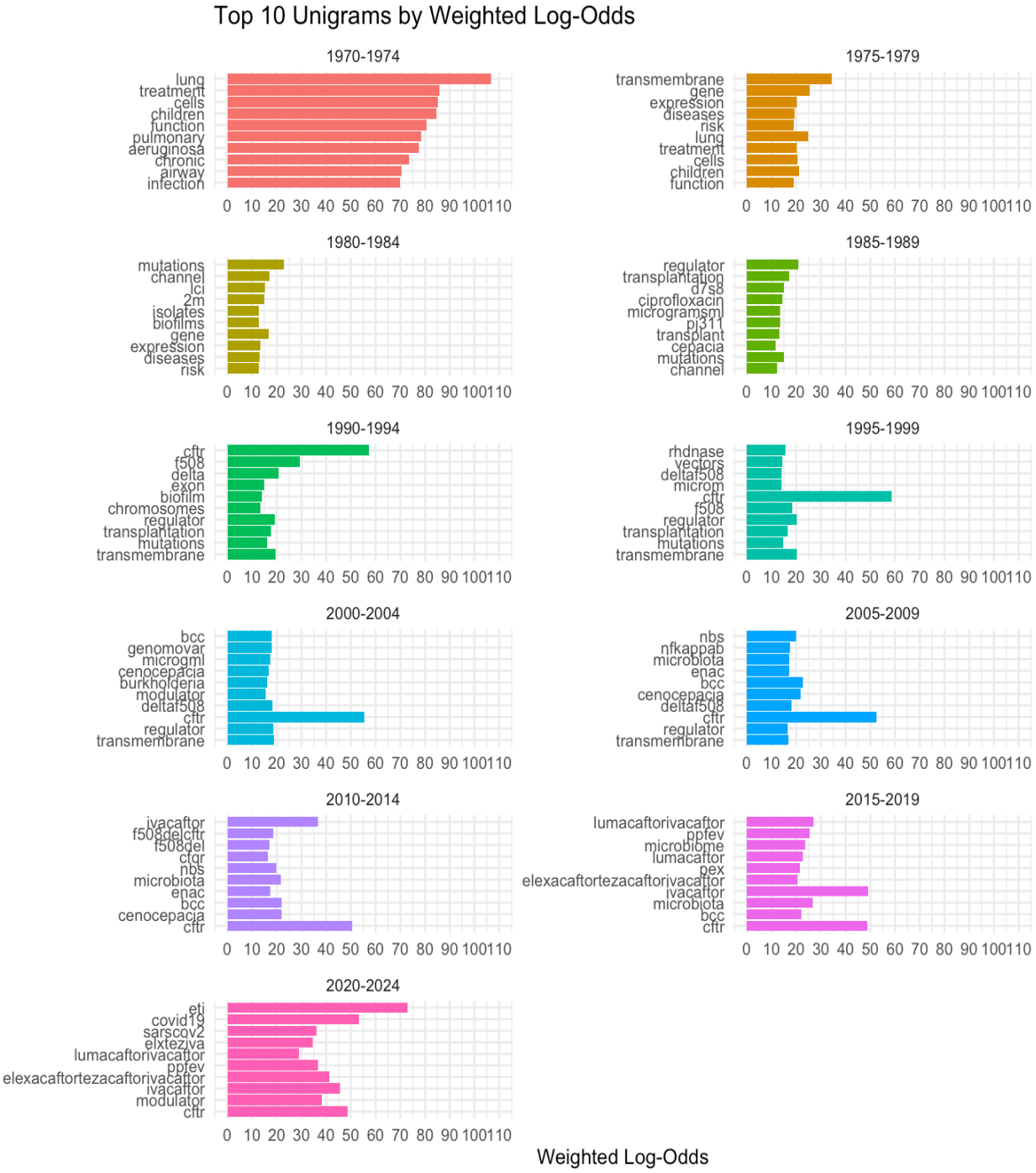


Figure 1. Top 10 unigrams by weighted log-odds for each quinquennial period from 1970-2024. Weighted log-odds correct for uneven corpus size across periods. Terms shown follow exclusion of disease-name and systematic review and methodology terms.



Figure 2. Top 10 bigrams by weighted log-odds for each quinquennial period from 1975-2024, following the same filtering process as Figure 1.



Figure 3. Top 10 trigrams by weighted log-odds for each quinquennial period from 1975-2024, following the same filtering process as Figures 1 & 2.

3.3 Vocabulary Richness

Table 2 presents vocabulary richness metrics for each quinquennial period. Total vocabulary size (*V*) grew from 1,372 in 1970-1974 to 40,880 by 2020-2024, while the Gini coefficient increased over the same period, from 0.479 to 0.873 (Figure 4), indicating that word usage became increasingly concentrated around a smaller set of dominant terms as the corpus grew.

The proportion of the vocabulary made up of hapax legomena fell sharply, from 59.5% in 1970-1974 to about 41-43% through the 1980s-2000s, before rising slightly to 44.1% by 2020-2024. This small increase coincides with the introduction of substantial new modulator and outcome-specific terminology identified in Section 3.2.

Table 2. Vocabulary richness metrics by quinquennial period, 1970-2024. *N*: total unigram tokens; *V*: unique unigram types; Hapax Legomena: words occurring exactly once; % Hapax: hapax legomena as a percentage of *V*; Gini: Gini coefficient; Yule's *K*: lexical repetitiveness statistic.

Vocabulary Richness Metrics by Quinquennial Period

Period	N (Tokens)	V (Unique Types)	Hapax Legomena	% Hapax	Gini	Yule's K
1970-1974	3,306	1,372	816	59.5	0.479	37.33
1975-1979	39,135	6,560	3,052	46.5	0.702	34.09
1980-1984	69,798	9,138	3,941	43.1	0.738	30.98
1985-1989	131,826	13,223	5,444	41.2	0.774	25.43
1990-1994	228,431	18,326	7,575	41.3	0.805	21.12
1995-1999	326,367	22,931	9,553	41.7	0.822	18.57
2000-2004	413,849	26,310	10,913	41.5	0.835	17.90
2005-2009	510,765	30,666	13,103	42.7	0.846	17.83
2010-2014	672,221	36,076	15,503	43.0	0.861	16.98
2015-2019	773,225	40,170	17,951	44.7	0.869	16.88
2020-2024	844,464	40,880	18,028	44.1	0.873	15.60

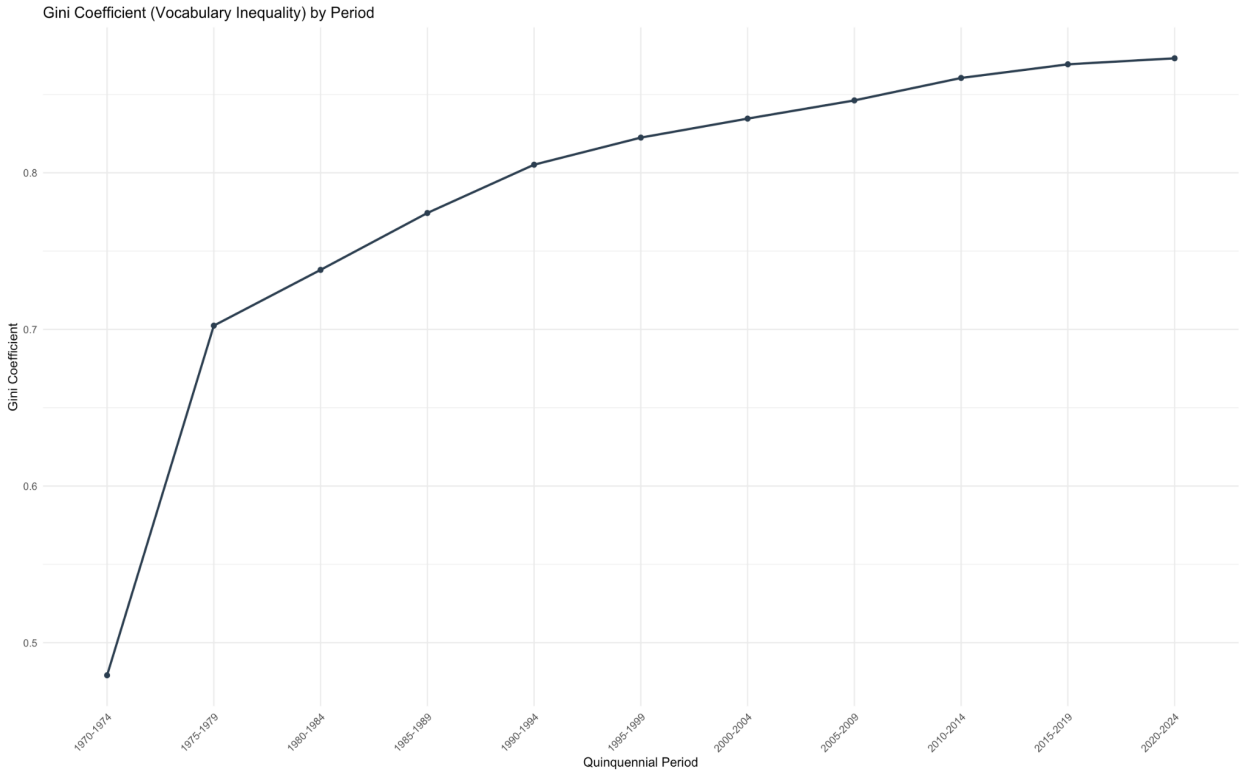


Figure 4. Gini coefficient of unigram frequency distribution for each quinquennial period, 1970-2024. Values near 0 indicate word usage spread evenly across the vocabulary; values near 1 indicate usage concentrated among a small number of dominant terms.

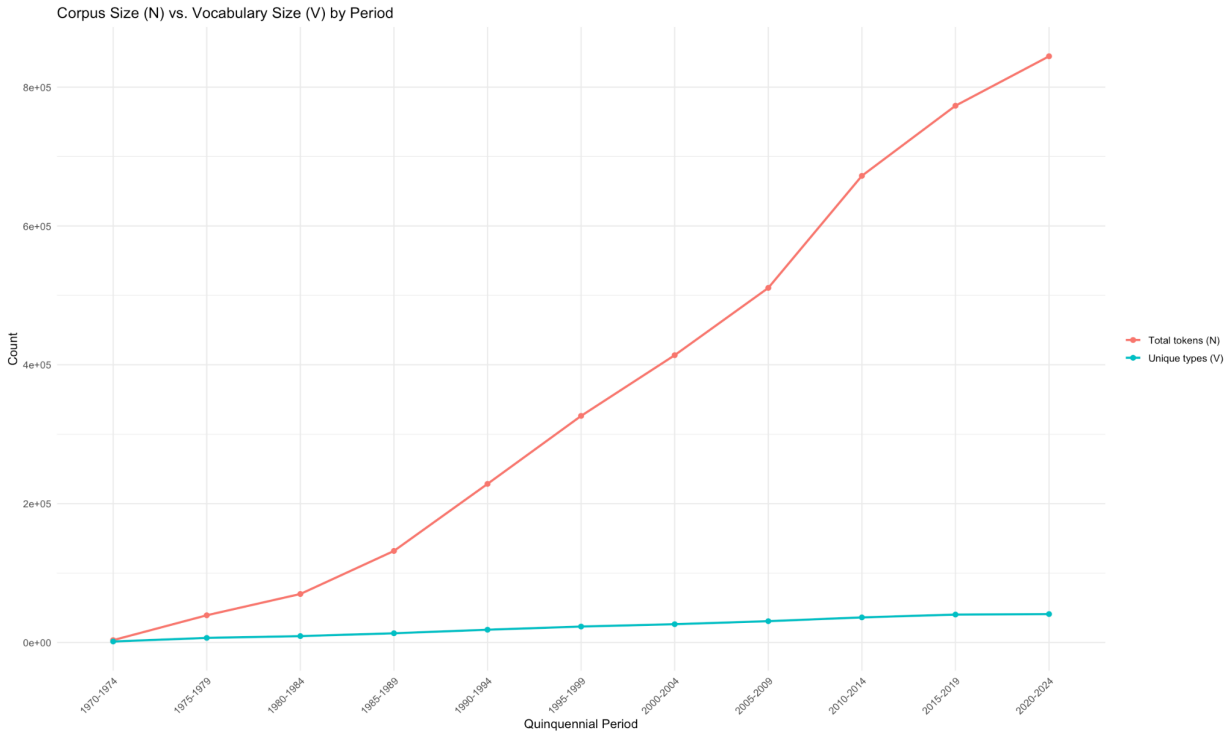


Figure 5. Total token count (*N*) and unique vocabulary size (*V*) by quinquennial period, 1970-2024.

Yule's *K* declined over the same interval, from 37.33 to 15.60 (Figure 6), indicating a simultaneous increase in overall lexical diversity.

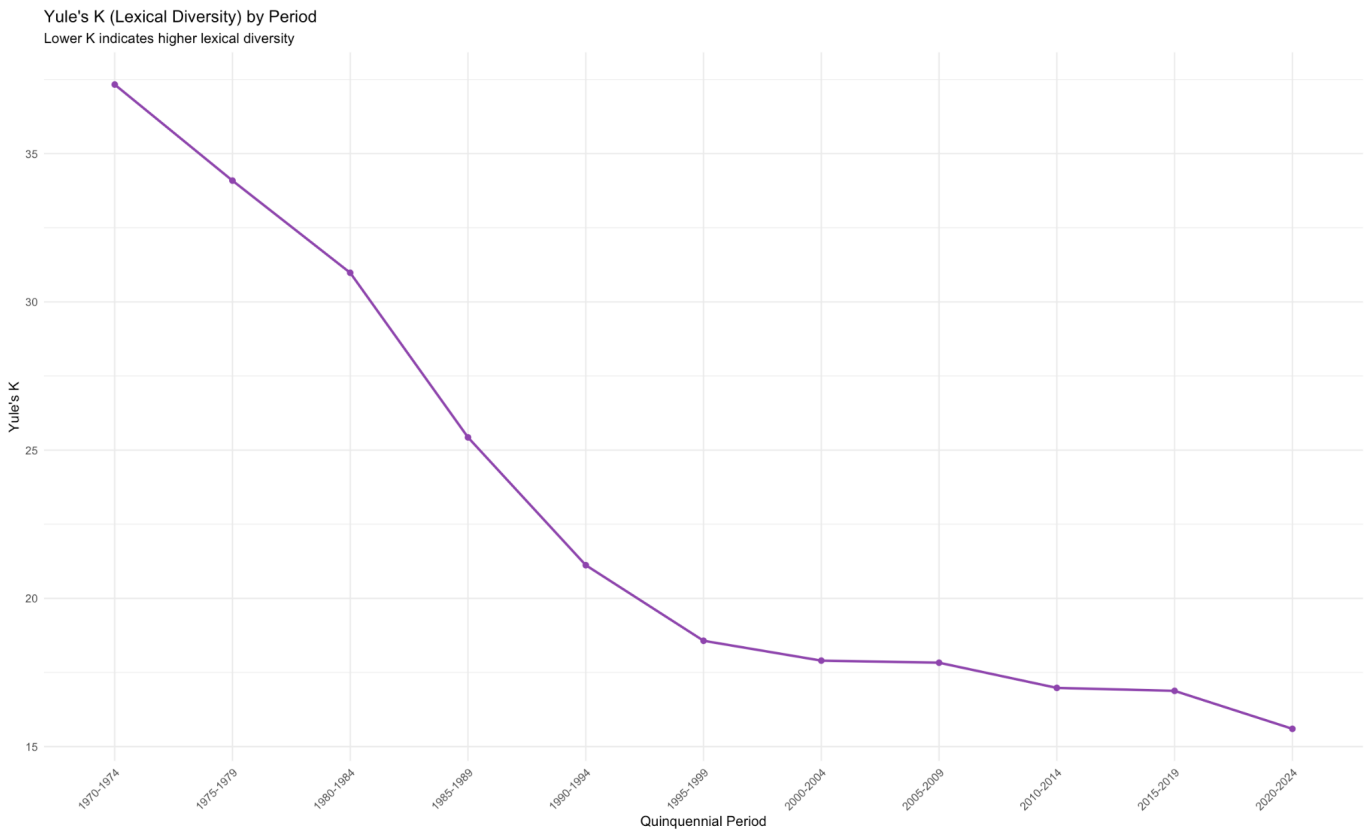


Figure 6. Yule's *K* statistic by quinquennial period, 1970-2024. Lower values indicate greater lexical diversity; higher values indicate greater repetitiveness.

Figure 5 shows that Gini rising and Yule's *K* falling are not actually contradictory. Both total word count (*N*) and vocabulary size (*V*) grew substantially over time, but in the early decades, *N* grew much faster than *V*. This means the same words were being increasingly repeated, rather than new words entering the vocabulary at the same pace. This gap between *N* and *V* narrowed in later periods, as vocab growth kept closer pace with the growth in total word count.

3.4 Relationship Between Term Distinctiveness and Vocabulary Concentration

To check whether periods with more distinctive terminology also had more concentrated vocabulary, each period's median log-odds score (from its 20 most distinctive terms) was compared against the period's Gini coefficient (Table 3), using Spearman's rank correlation. The results differed depending on the type of term:

- **Unigrams**: a weak positive correlation ($\rho = 0.300$, $p = 0.371$); not statistically significant

- **Bigrams**: a moderate negative correlation (ρ = -0.576, p = 0.088); close to being significant, but not quite reliable
- **Trigrams**: a strong negative correlation (ρ = -0.758, p = 0.016); statistically significant

The strength and direction of the correlation shifted with n-gram length, which moved from a weak positive relationship for unigrams to a more significant negative relationships for trigrams.

Table 3. Spearman correlation between median weighted log-odds (top 20 terms) and Gini coefficient by period, for unigrams, bigrams, and trigrams.

Spearman Correlation Between Median Weighted Log-Odds and Gini Coefficient

Top 20 terms by period, per n-gram type

N-gram Type	Spearman's rho	*p*-value	Significant ($p < .05$)
unigram	0.300	0.371	FALSE
bigram	−0.576	0.088	FALSE
trigram	−0.758	0.016	TRUE

3.5 Treatment/Adverse Event Co-occurrence Network

A co-occurrence network was built using a manually chosen set of treatment and adverse-event terms, based on the original search terms and supplemented with a small number of additional terms found to strongly co-occur with them. In this kind of network, each word or phrase is a "node" and a line (or "edge") connects two nodes whenever they appear together in the same abstract. The resulting network (Figure 7) contained 73 nodes and 366 edges, where 10 nodes were treatment terms, 7 were adverse event terms, and 56 were other related terms that appeared alongside them.

The network formed into several distinct clusters. A CFTR modulator cluster (ivacaftor, lumacaftor, tezacaftor, elexacaftor, cftr modulator) connected to CFTR-related biological terminology (transmembrane conductance regulator, F508del) and to general adverse-event/safety terminology (safety, adverse events, adverse effects). A separate cluster centred on tobramycin connected to infection and pathogen-related terminology (*Pseudomonas aeruginosa*, infection, chronic). Mental-health terms (anxiety, depression, mental health) formed a further, separate cluster, with no edges connecting them directly to the CFTR modulator cluster at the co-occurrence threshold used to construct this network.

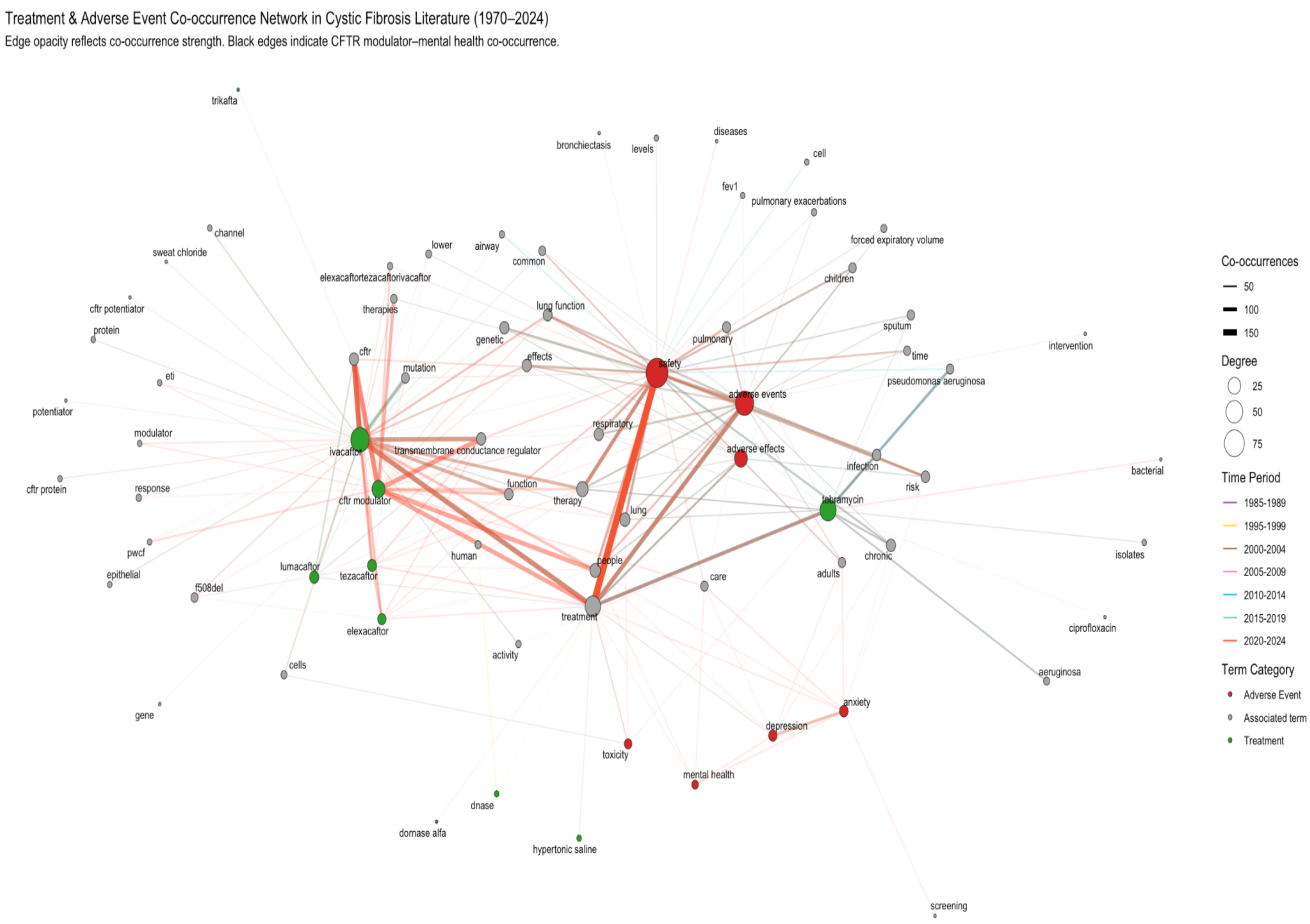


Figure 7. Co-occurrence network of treatment and adverse-event terminology in CF literature from 1970-2024. Node colour indicates category (Treatment, Adverse Event, or Associated Term); node size reflects degree. Edge colour indicates the quinquennial periods of co-occurrence; edge opacity and width scale with co-occurrence frequency.

3.6 Extent of Modulator/Mental Health Co-occurrence

The main network graph (Section 3.5) only shows connections strong enough to clear a minimum threshold. To get a fuller picture of how any potential CFTR modulator/mental-health link developed over time, I also investigated the raw, unfiltered co-occurrence counts, with no threshold applied (Figure 8).

Given prior literature specifically linking CFTR modulator therapy to mental health effects (Piehler et al., 2023), and the rise in both modulator and mental-health-related language identified elsewhere in this analysis (Sections 3.2, 3.7), a more substantial co-occurrence between these two term categories by 2020-2024 was anticipated. However, the results did not support this assumption. No CFTR modulator term and mental-health term appeared together in the same abstract at all before 2015-2019. By 2020-2024, every examined pair had co-occurred at least once, but the strongest pair (mental health/CFTR modulator) reached a maximum of only 8 co-occurrences. For comparison, "anxiety" (n=340), "depression" (*n*=275), and "cftr

modulator" (*n*=436) each individually appeared several hundred times across 2020-2024 abstracts, meaning a measly percentage of "mental health" mentions in this period occurred in the same abstract as a CFTR modulator term.

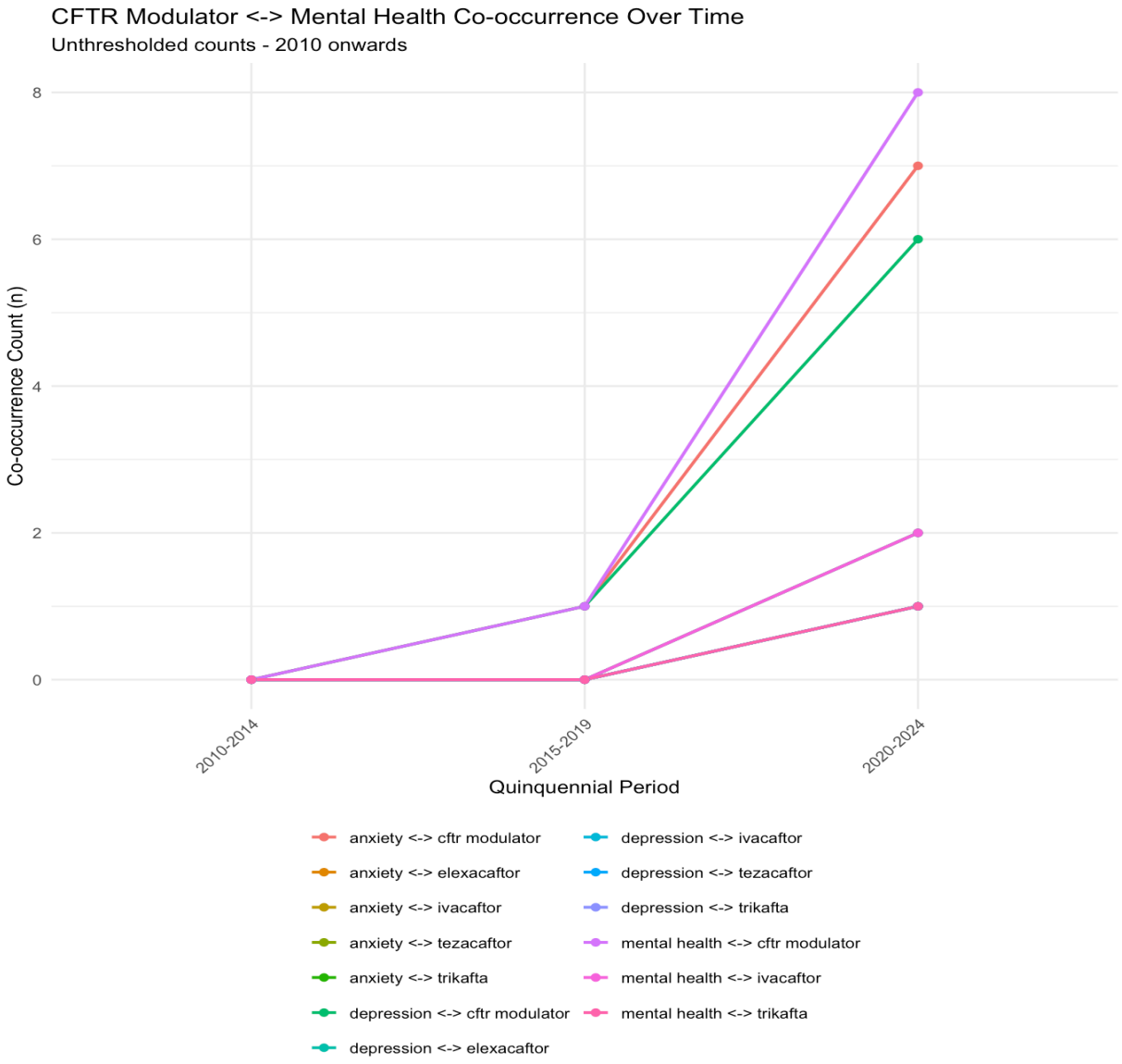


Figure 8. Co-occurrence between CFTR modulator-family and mental-health-related terms by quinquennial period from 1970-2024, using non-thresholded counts.

3.7 Period-over-Period Change in Term Usage

The average term rate change from one period to the next was measured separately for treatment terms and adverse-event terms (Figure 9). Treatment-term usage rose sharply starting in 2005-2009, around the same time modulator drugs began appearing among the most distinctive terms of 2010-2014.

Adverse-event-term usage showed its biggest jump over that same 2010-2014 period. However, the most distinctive terms for that specific period were mostly about biology and clinical outcomes, not adverse-event reporting.

Treatment-term usage dipped slightly in 1990-1994 and 2000-2004, meaning the average rate briefly fell compared to the period before. Adverse-event-term usage only slightly dipped once in 1990-1994.

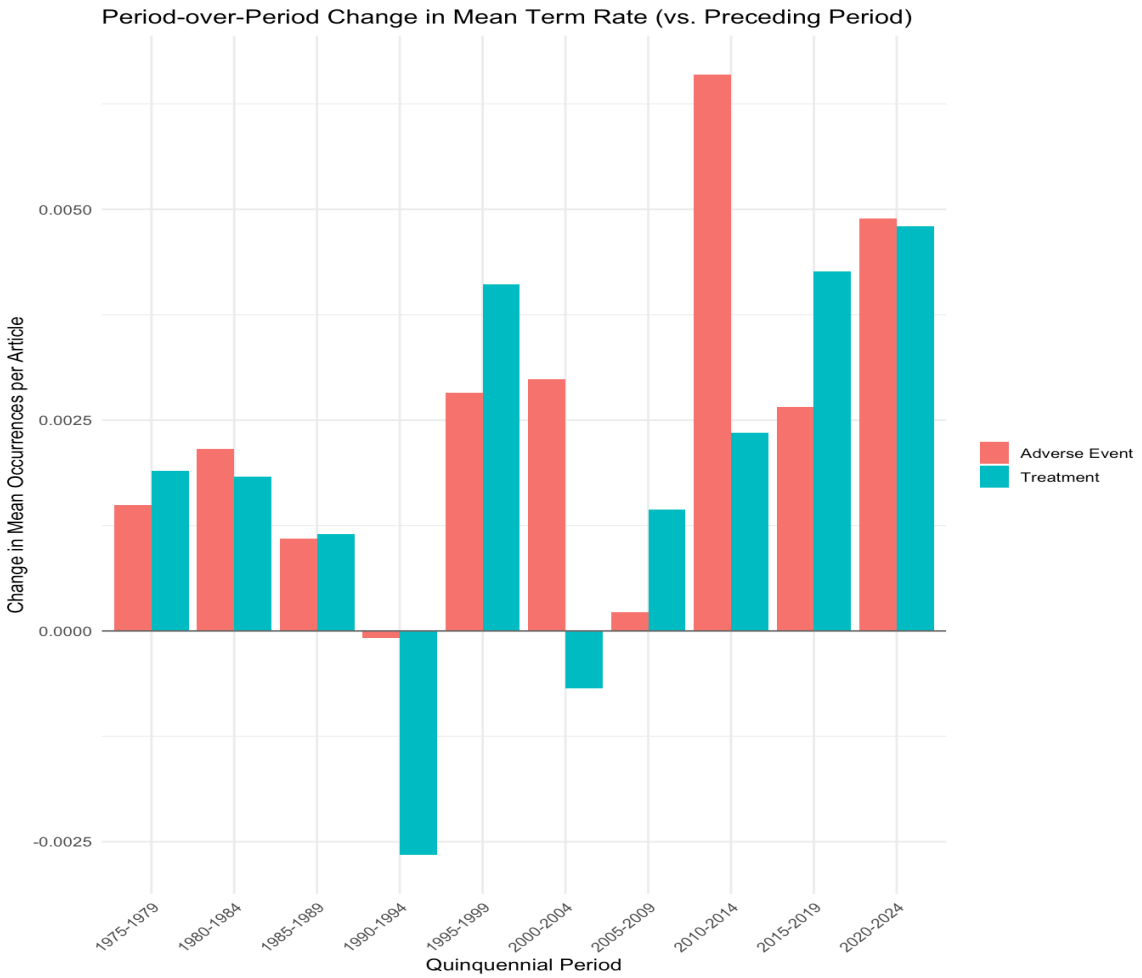


Figure 9. Period-over-period change in mean per-article occurrence rate for fixed treatment and adverse-event terms, relative to the immediately preceding period.

Discussion

4.1 Evidence of a Genuine Treatment-Era Terminology Shift

The shift is clear, as we move from the earliest periods to more contemporary periods. The field of CF's most distinct terms have shifted from generic symptom terms in the 1970s toward molecular language following the CFTR gene's discovery in 1989, likely reflecting both the

field’s move from describing symptoms to understanding CF at a molecular level. The small size of the early corpus likely limited distinct terminology from appearing. Several different analyses, using different methods, all point to the same clear finding: CF literature underwent a real shift in language, coinciding with the introduction of CFTR modulator therapy. The period-specific log-odds analysis (Section 3.2) shows language moving from general symptom-related and genetic terms in earlier decades toward modulator drugs and lung-function measures by 2020-2024. This is backed up by the fixed-term rate trends (Section 3.7), where "ivacaftor" and "cftr modulator" both show sharp increases in the most recent periods, and by the co-occurrence network (Section 3.5), where the CFTR modulator cluster is one of the most tightly connected parts of the whole network. The fact that four different methods all point to the same shift makes it much more likely this is a genuine change in the literature, rather than something caused by one analytical choice.

The timing of this shift also matches what is known clinically. The sharpest rise in modulator-specific language begins in 2010-2014, matching ivacaftor's licensing approval in 2012 following pivotal clinical trials (Ramsey et al., 2011; Deeks, 2013) and grows even further by 2020-2024, matching the introduction of combination modulator therapies (Hoy, 2019). This match between a real, independently known clinical timeline and a pattern found purely from text gives extra support that the text-mining approach used is picking up genuine signal.

4.2 The Treatment-Adverse Event Relationship

A aim of this project was to map how treatment and adverse-event language relate to each other. The network analysis (Section 3.5) shows that this connection does exist. CFTR modulator terms link, through a small number of edges, to mental-health-related adverse-event terms, and more broadly, adverse-event/safety language connects to multiple treatment clusters (both modulators and tobramycin).

However, the supplementary analysis using unfiltered counts (Section 3.6) shows this connection is weaker and more recent than the network graph alone might suggest. No CFTR modulator term and mental-health term appeared together at all before 2015-2019, and even by 2020-2024, a negligible amount of mental-health-term mentions appeared alongside a modulator term in the same abstract. It is also important to consider the potential for a lag period between the introduction of a treatment, its side effects, and follow up reporting through literature. Therefore, change in adverse event patterns may take time to become detectable.

This matters for how the network graph should be read. Looking at Figure 7 on its own, someone might assume modulator/mental-health links are a well-established, common feature of CF literature, but Section 3.6 shows this isn't quite right. The connection is better described as more of a developing relationship. Even though earlier clinical research specifically flagged modulator-related mental health effects (Piehler et al., 2023), the text-mining evidence here suggests this link is not yet a prominent part of how treatment and adverse events are discussed together, at least in wider literature. This approach also can’t determine if literature is reporting a negative or positive association of anticipated mental health effects. This is important due to the potential relevance of mental health effects associated with CFTR modulators and could be

further contextualised by regulatory concerns, such as the MHRA alert relating to Kaftrio (UK Cystic Fibrosis Registry, 2025)

A further limitation came from the period-over-period analysis (Section 3.8). The biggest single jump in adverse-event-term usage (from 2005-2009 to 2010-2014) happened at the same time modulator language was entering the literature. But the most distinctive terms for that specific period (Section 3.2) were mostly about biology and clinical outcomes, not adverse-event reporting itself. This shows that just because treatment language and adverse-event language rise at the same time doesn't necessarily mean they're directly linked.

4.3 Vocabulary Concentration and Diversification

The vocabulary richness findings (Section 3.3) show a pattern that looks contradictory at first but makes sense once explained: the Gini coefficient rose while Yule's *K* fell, meaning CF literature's vocabulary became more concentrated around a core set of dominant terms and more lexically diverse overall, at the same time. This fits with the CF research field, where a small set of concepts (CFTR, modulator) get repeated more often, even as the field branches out into more specialised sub-areas, each bringing its own set of rarely repeated terms.

The correlation analysis (Section 3.4) adds an important caveat to take note of. A significant negative relationship between median log-odds and Gini coefficient showed up for trigrams, but not for unigrams or bigrams. This suggests that at least some of the apparent link between vocabulary concentration and term distinctiveness is just a byproduct of how log-odds calculations behave on larger samples, rather than a genuine finding about the language itself. It shows that summary statistics calculated from a corpus of very different sizes need to be interpreted carefully, and that findings shouldn't be too quickly attributed to real changes in content without first considering how the growing size of the corpus itself affects the underlying statistics.

4.4 Methodological Reflection

A few decisions made during this analysis have a direct effect on how the results should be interpreted. Most importantly, the network vocabulary was ultimately built with clinical insight from my supervisor and awareness of common phrases and abbreviations within CF literature, rather than the purely data-driven, log-odds-ranked approach that was first attempted. This change was necessary because the terms that scored highest as "statistically distinctive" were often not actually about treatment or adverse events. Often, systematic-review terms and generic academic language kept dominating the results, even after extensive filtering. The hand-picked approach produces a clearer, more directly relevant network, but it also means the analysis can't discover treatment or adverse-event terms that weren't already anticipated when the search terms were designed. The validated data-driven expansion helps address this gap, but only seven additional terms passed the strict validation criteria used. With more time and resources, this vocabulary could likely be expanded further.

A second trade-off worth noting is the decision to keep stopwords within bigrams and trigrams, to preserve grammatical context. This meant common bigrams and trigrams in the corpus (like

"in the" and "of the") had to be removed from the log-odds analysis through a separate filtering step (Section 2.6), rather than simply not being generated in the first place which made it the pipeline increasingly more complex. Removing stopwords earlier, at the tokenisation stage, would have made this simpler, but at the cost of losing genuine phrase context.

4.5 Limitations

This analysis has several limitations. First, using titles and abstracts rather than full-text articles limits the depth of relationships that can be identified. More nuanced or directional claims (for example, a treatment reducing versus causing an adverse event) cannot be distinguished through co-occurrence alone. Second, despite an extensive filtering process, a small number of artefacts remained in the results. Some were character encoding issues, and others were systematic review phrasing that hadn't yet been added to the exclusion criteria. An example is seen in Figure 3 where the "2016 wiley periodicals" (2015-2019) still appears. This shows a general limitation of the list-based filtering approach used since it can only exclude wording it has explicitly seen in the code. Therefore, new phrasing of an already excluded concept can still slip through. More iterative review or some form of pattern-based filtering approach could possibly solve this issue. Third, English-language restriction, while necessary for consistent n-gram analysis, may introduce geographic bias, and advanced therapies are less accessible to low-income countries and middle-income countries (Guo et al., 2024). This could potentially underrepresent CF research conducted in non-English-speaking regions. Fourth, there are other scientific databases that could be explored in addition to PubMed to create a more thorough, comprehensive usage of textual data. Fifth, the search strategy did not filter by article type, so both original research and review articles were included. This could explain some of the unexpected terms found in the top-ranked terminology. Although restricting the corpus to original research only could reduce this kind of noise, it would also lose useful content from review articles. Finally, the curated network vocabulary (Section 4.4) reflects the treatment and adverse-event categories anticipated at the start of this study. So, while a validated data-driven expansion step was applied, some genuine associations may remain undetected simply because the relevant terminology was not included in the search vocabulary from which the network was made.

Overall, this study illustrates text-mining can be useful for identifying patterns across a large body of scientific literature that would be difficult to identify through manual review alone. Although this analysis focused on CF, the same general approach and techniques could be applied to other diseases to investigate how terminology and other associations between different concepts change over time.

Ethical Issues

The dataset used (PubMed) is a publicly available data source. No patient data, personally identifiable information, or confidential material is used. Additionally, the study did not require any ethical approval as there are no human participants in this project.

AI Statement

AI-based tools (Claude, ChatGPT, Grammarly) were used in an assistive role only, limited to:

- Improving sentence clarity and checking grammar and spelling
- Supporting literature review and suggesting relevant citations for methodological steps
- Scanning data frames to identify duplicate terms or variant spellings/forms of the same term

All written work, coding, analysis, methodology, and interpretation are the student's own. AI-generated suggestions were critically reviewed for accuracy before use, and no AI tool was used to generate data, conduct analysis, or produce results.

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Appendix

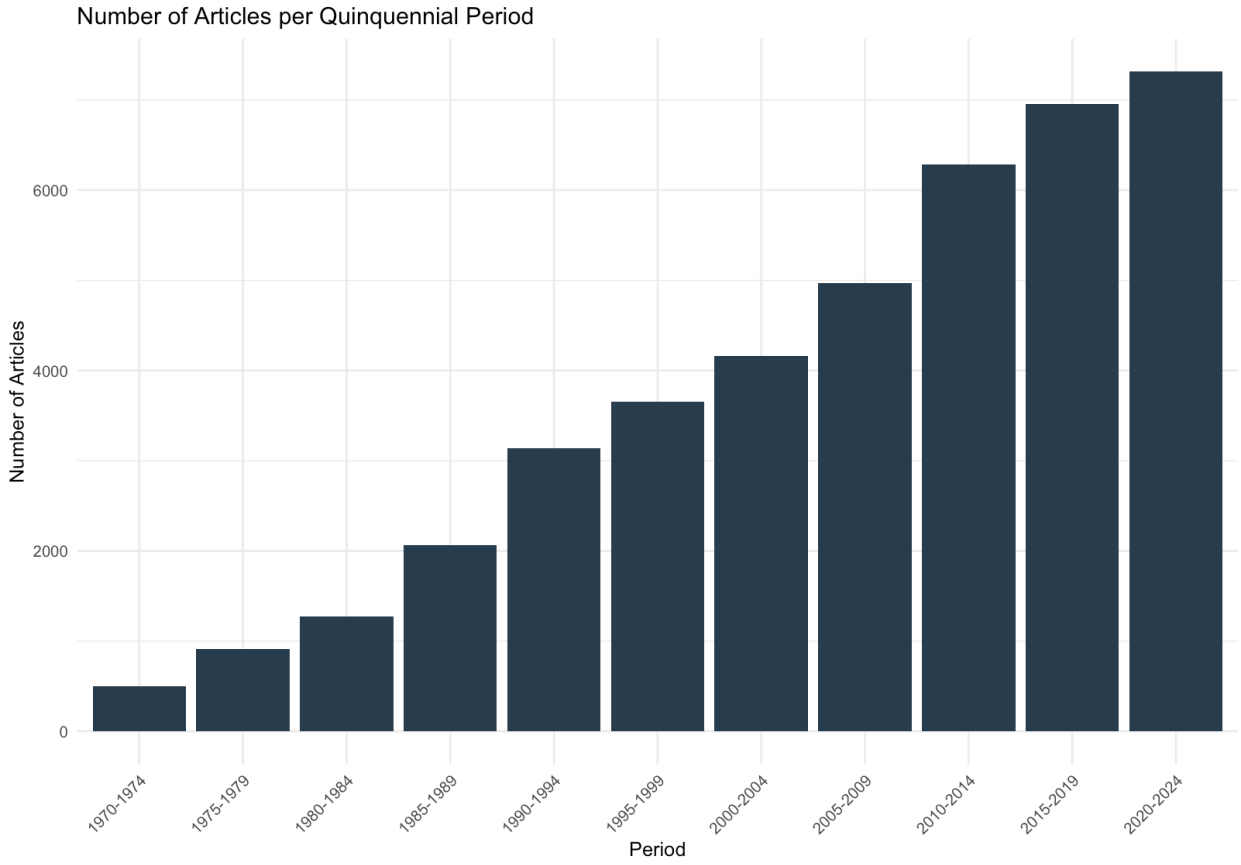


Table (Appendix). Number of unique articles retrieved per quinquennial period, following deduplication on PMID (N = 41,222).